

JOSEPH ADEOLA

+34 610-423-930 2595GA, The Hague, Netherlands
adeola.jo@outlook.com LinkedIn
GitHub Portfolio

Profile

Robotics researcher and software developer with a background in mathematics and intelligent robotic systems. Currently pursuing a PhD at TU Delft in collaboration with NXP Semiconductors, developing methods for acoustic object localization and multimodal perception in autonomous driving. Previous work includes C++ coordination software for a four-robot medical imaging system and deep learning pipelines for image analysis.

Research & Professional Experience

Sept 2025 – Present

TU Delft, Intelligent Vehicles Group, Netherlands

Doctoral Researcher, Department of Cognitive Robotics, in collaboration with NXP Semiconductors

- Developing uncertainty-aware deep learning approaches to acoustic object localization in bird's-eye view (BEV), alongside research on fusion with lidar, camera, and radar for autonomous driving.
- Developing and maintaining components of the group's localization stack, covering map building, multi-sensor calibration, data collection campaigns, and SLAM pipelines.
- Supervising four bachelor end-project students, one research intern, and one master's student on acoustic perception for autonomous driving, including in-cabin microphone array placement in a Toyota Prius research vehicle.
- Guiding MSc students through labs and course projects in Machine Perception for Robotics and Robot Software Practicals, and assessing their work as a Teaching Assistant.
- Chairing weekly Intelligent Vehicles group meetings and coordinating invited research talks on perception, localization, control, and teleoperation, with speakers from European and Asian universities.

Oct 2024 – Mar 2025

ViCOROB Institute, Girona, Spain

Senior Technician

- Developed C++ distributed-control software and real-time coordination modules for a four-robot system in *iToBoS*, an EU Horizon 2020 project for early melanoma detection through automated total-body imaging.
- Implemented computer vision and machine learning algorithms in Python and OpenCV for detection and classification in the imaging workflow.
- Developed sensor fusion algorithms to connect machine learning models with robotic control systems.
- Contributed to research publications on automated imaging and detection, alongside the project's software development work.

June 2023 – July 2024

■ **ViCOROB Institute, Girona, Spain**
AI Research Intern

- Implemented deep learning models for object detection and classification, supporting the development of AI-based imaging systems.
- Managed research databases and automated data cleaning to prepare datasets for machine learning development.
- Built preprocessing pipelines that improved model accuracy by 5%.
- Applied SHAP, layer-wise relevance propagation (LRP), and Grad-CAM to explain model predictions and inspect the image regions driving classification.
- Worked with cross-functional teams to integrate trained models into existing software systems.

Feb 2022 – Sept 2022

■ **KPMG, Lagos, Nigeria**
Financial Risk Management Intern

- Implemented a stock option pricing program in C++ and validated its outputs against market data, improving risk assessment accuracy by 15%.
- Developed Probability of Default (PD) and Expected Credit Loss (ECL) tools in Python and R, reducing analysis time by 30%.
- Built financial models in R, Excel, and VBA to analyze market trends and generate forecasts.
- Supported risk assessment engagements for five Nigerian banks against Central Bank of Nigeria requirements.

June 2021 – Nov 2021

■ **University of Ilorin, Department of Physics, Nigeria**
Graduate Research Assistant

- Developed Python algorithms and Windows software to automate research data cleaning, increasing processing efficiency by 40% and reducing manual errors by 60%.
- Collected, cleaned, and managed research datasets in Python and Excel, achieving 99.9% data accuracy and completeness.
- Analyzed research data and contributed findings to journal publications.
- Assisted in designing and implementing experimental protocols for research data collection and analysis.

Education

Sept 2025 – Present

■ **Doctoral Candidate in Cognitive Robotics**
TU Delft, Netherlands

2024 – 2026

■ **MSc coursework in Financial Engineering**
WorldQuant University, programme not completed.

Sept 2022 – June 2024

■ **Erasmus Mundus Masters, Intelligent Robotic Systems**
University of Zagreb, Croatia & University of Girona, Spain
[Grade: Excellent \(9.05/10\)](#) | [Course transcript](#)

Oct 2016 – Aug 2021

■ **Bachelor of Science in Mathematics**
University of Ilorin, Nigeria
[First-Class Honors \(4.92/5.0\)](#); University Valedictorian

Summer Schools

20–26 June 2026

■ **ETH Robotics Summer School**, ETH Zürich, Switzerland

15–29 June 2025

■ **Summer School in Physics-Informed Neural Networks**, KTH Royal Institute of Technology, Sweden

Publications & Manuscripts

Journal Article

- 2025  Saha, A., **Adeola, J.**, Ferrera, N., et al. Skin region images extracted from 3D total body photographs for lesion detection. *Scientific Data*, 12, 1442. [DOI](#)

Preprint

- 2022  Bello, S.A., Yusuf, K.A., Ige, S.O., **Joseph, O.A.**, et al. DINData: A Windows software to reformat and clean digisonde numeric dataset. *ESS Open Archive*.

Manuscript Under Review

-  Ziang Liu, Shiming Wang, **Joseph Adeola**, and Julian Kooij.
AcousticBEV: Acoustic BEV Object Localization for Intelligent Vehicles.
Under review at *IEEE Robotics and Automation Letters* (RA-L).

Technical Skills

Programming	 Python, C++; MATLAB, R, C, Rust, JavaScript
Robotics & perception	 ROS/ROS2, PCL, Gazebo, OMPL, PDDL; sensor fusion, SLAM, mapping, obstacle detection, motion planning, control systems
Machine learning	 PyTorch, TensorFlow, Keras, scikit-learn, OpenCV; object detection, classification, tracking, CNNs, vision transformers
Acoustic processing	 Audio signal processing, microphone arrays, STFT, spectrogram analysis, beamforming, audio classification, sound event detection
Data & analysis	 NumPy, Pandas, R, Excel; data wrangling, statistical analysis, visualization
Development tools	 Git, Docker, Linux/Ubuntu, LaTeX

Projects

-  Full portfolio: [GitHub Projects Page](#).

Robotic Perception & Sensor Fusion

-  **Autonomous Exploration Framework:** Built an end-to-end robot exploration system that combines sensor fusion, mapping, and navigation. [Demo](#)
-  **Stereo Visual Odometry:** Implemented feature tracking, outlier rejection, and motion estimation on KITTI sequences to estimate camera motion. [Code](#)
-  **Camera Calibration:** Built a Python/OpenCV pipeline to estimate intrinsic and extrinsic camera parameters. [Code](#)
-  **Event-Based Feature Tracking:** Implemented event-based feature tracking using the iterative closest point (ICP) algorithm, targeting challenging lighting conditions. [Demo](#) | [Report](#)
-  **iToBoS Annotation Tool:** Built a desktop application in React, Electron, and Vite for creating YOLO-format annotations for machine learning datasets. [Code](#)

SLAM

-  **Pose-Based EKF SLAM:** Combined extended Kalman filtering with ICP scan matching to estimate robot pose and build a map on a Kobuki Turtlebot. [Code](#) | [Report](#)
-  **EKF Map-Based Localization:** Implemented extended Kalman filter localization using line features and a measurement model for robot pose estimation. [Code](#) | [Report](#)
-  **Particle Filter Localization:** Developed particle filter localization for a Kobuki Turtlebot, using adaptive resampling to maintain a probabilistic estimate of robot pose. [Code](#) | [Report](#)

Robotics & Autonomous Systems

- **Multi-Robot Swarm Control:** Implemented Reynolds' flocking rules in ROS and Gazebo to coordinate the motion of a robot swarm. [Code](#)
- **Multi-Modal Human-Swarm Interaction:** Integrated vision, speech recognition, and large language models into a multimodal interface for human interaction with a robot swarm. [Code](#)
- **Path Planning & Benchmarking:** Implemented and benchmarked A*, RRT, RRT*, and RPS to compare alternative algorithms for robot path planning. [Code](#)
- **Multi-Robot Gazebo Spawning:** Developed a ROS package with parameterized configurations for spawning multiple robots in Gazebo simulations. [Code](#)
- **Quadcopter Control System:** Designed PID controllers for roll, pitch, yaw, and altitude, with an interface for tuning gains in real time. [Code](#)

Machine Learning, Object Detection & Classification

- **Skin Lesion Detection:** Compared CNN and vision transformer models for clinical skin lesion image classification, achieving 89% accuracy. [Code](#)
- **CNN-Based Image Classification:** Compared ResNet, VGG, DenseNet, and custom CNN architectures for image classification with limited training data. [Code](#)
- **Deep Learning Sentiment Analysis:** Implemented LSTM, RNN, and GRU models for sentiment classification, achieving 91% accuracy. [Code](#)
- **Spiking Neural Networks for Classification:** Implemented and compared spiking neural network architectures for handwritten digit classification on Neuromorphic MNIST. [Code](#)
- **Graph Neural Networks for Citation Analysis:** Implemented GCN, GAT, and GraphSAGE models to classify research papers and predict missing citation links in the Cora dataset. [Code](#)
- **Privacy-Preserving Federated Learning:** Built a federated learning framework incorporating differential privacy for distributed model training. [Code](#)
- **Contrastive Metric Embedding:** Implemented unsupervised metric embedding and similarity learning for image classification with limited labeled data. [Code](#)
- **Reinforcement Learning for Motion Planning:** Implemented Q-learning for navigation, with support for transfer learning. [Code](#)

Technical Reports

- 1 Enhancing Autonomous Exploration and Navigation in Unknown Environments: A Software Architecture for Turtlebot Robot - **Joseph Adeola**, Preeti Verma, and Moses Ebere - [Demo](#) | [Report](#)
- 2 Controlling a Swarm of Robots in a Simulator using Reynolds' Rules - **Joseph Adeola**, Khawaja Ghulam Alamdar, Moses Chuka Ebere, Nada Elsayed Abbas - [GitHub](#) | [Report](#)
- 3 Event-Based Feature Tracking Using the Iterative Closest Point Algorithm - **Joseph Adeola**, Preeti Verma, and Moses Ebere - [Demo](#) | [Report](#)
- 4 Multi-Modal Human-Swarm Interaction Using Vision, Speech & Multi-Agent LLMs - **Joseph Adeola**, Moses Ebere - [GitHub](#) | [Report](#)
- 5 Robust Coordination and Control of Multi-Robot Systems Using Consensus Protocols - **Joseph Adeola**, Khawaja Ghulam Alamdar, Moses Ebere, Nada Elsayed Abbas - [GitHub](#) | [Report](#)
- 6 Pose-Based EKF SLAM using ICP Scan-Matching on a Kobuki Turtlebot - **Joseph Adeola**, Moses Ebere, and Preeti Verma - [GitHub](#) | [Report](#)
- 7 A Kinematic Control System for a Mobile Manipulator, based on the Task-Priority Redundancy Resolution Algorithm - Moses Ebere, **Joseph Adeola**, and Preeti Verma - [Report](#)
- 8 Pigmented Skin Lesion Detection in Clinical Images Using Deep Learning Methodologies - **Joseph Adeola** - [GitHub](#) | [Report](#)

Awards

- 2025  Joan Oró FPI Predoctoral Scholarship (4-year PhD Fellowship), Government of Catalonia.
- 2023  ITExperience Google Project Management Career Scholarship Award.
- 2022  Erasmus Mundus Joint Masters Degree Scholarship Award.
 Education USA (Nigeria) Scholarship Award.
- 2021  Valedictorian, University of Ilorin, 36th Convocation.
 Best Graduating Student, Faculty of Physical Sciences, University of Ilorin.
 Best Graduating Student, Department of Mathematics, University of Ilorin.
 University of Ilorin International Scholarship Forum Award.
 Scholar of the Year Award, National Association of Mathematics Students, University of Ilorin Chapter.
- 2019  Bronze Medalist, International Youth Mathematics Challenge.
 MTN Foundation Scholarship Award, MTN Nigeria.
- 2017–2020  University Senate Scholarship Award, University of Ilorin.

Professional Memberships

- Nov 2021 – Present  Member, Data Science Association
- May 2021 – Present  Member, Nigerian Mathematical Society
- July 2020 – Present  Member, Data Science Nigeria

Languages

- English  C2